Explicit degree-three covers and toy subgroup transfer

E19; 2026-10-23; local bounded evidence, L0.

The BHLS genus-two construction (arXiv:1403.6911, Algorithm 5.4 and Appendix A) produces four parameter presentations per source, with exact degree-three maps to the original curve and its twist. Presentations are not asserted nonisomorphic.

The three toy sources have actual divisor pullback and norm replay with norm composed with pullback equal to multiplication by three. Every source point, inverse and generator translation is checked. Eight distinct public targets per toy source are solved in the destination subgroup and verified back on the source; public P-521 has no DLP attempt.

Source	Source points / translation checks	Recovery attempts	Distinct targets
p=211	233	8	8
p=751	773	8	8
p=10111	10079	8	8

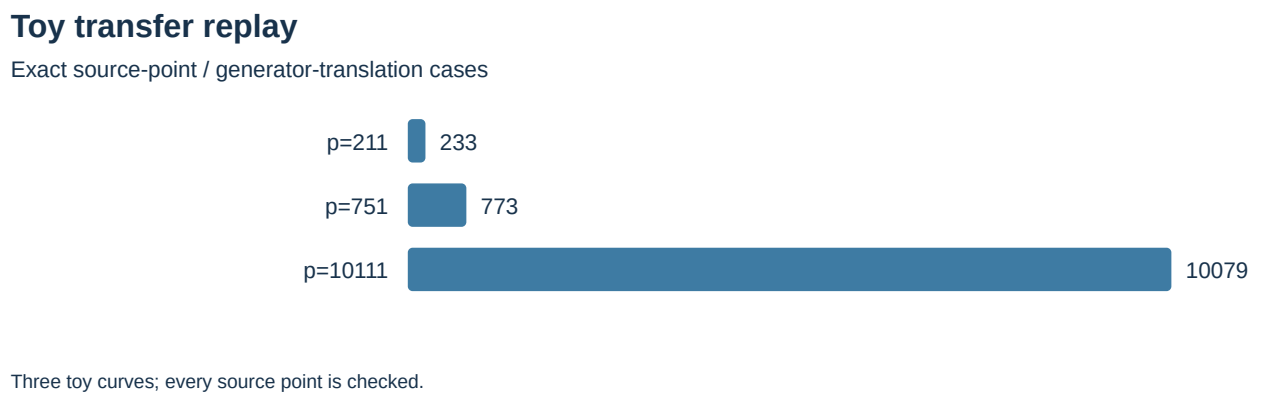

Figure 1: E19: exhaustive toy pullbacks; 24 distinct public targets across three source fixtures.

Polarized identification with each E18 graph, full second-map divisor transfer, balanced base-field arithmetic, formal chart/group bridges and all costs remain open. The construction is prior art, not a novelty claim.

Verified locally; complete raw publication awaits artifact-store access. No calibrated speedup, novelty, independent-host replay or NIST DLP is established.

Native assessment: ../../execution-v2/assessment-20261023-v3/ASSESSMENT.json. Raw members: runs/geometry-*, runs/three-transfer-20261023-v1 and -v2, and all CAS failures/replays. Source/input/binary hashes and exact argv: E19-receipts.json and ../../execution-v2/receipts/.

All phase receipts, including errors and timeouts: E19-receipts.json. The complete execution ledger and original protocols retain actual limits and missing measurements.